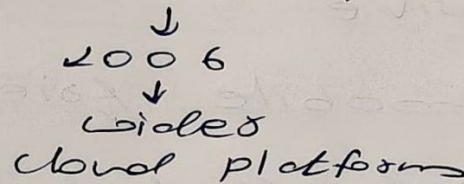


and computing is a technology
that provides various computing
resources such as servers, memory,
storage, etc. over the internet.

we need to "pay per use"

24x7 monitoring
natural disaster } → Traditional concepts

Google Drive, AWS, Microsoft Azure



• On-demand access :- Whenever a resource is needed, it can be accessed through cloud.

- Internet-based

① Private cloud

② Public cloud

③ Hybrid clonal

① > Private Cloud

only for one organization

* Characteristics

- ① Dedicated to a single organization
- ② High security and controls
- ③ Suitable for sensitive data

* Advantages

- ① Better data security
- ② Customization

* Drawbacks

- ① Expensive
- ② Not mobile friendly
- ③ Limited scalability

* Examples

- Private cloud in banks

② > Public Cloud

shared with multiple users

* Characteristics

- ① Open for public use and shared among users
- ② managed by service providers
- ③ Cost-effective

* Advantages

- ① Low cost
- ② Easy to use
- ③ No maintenance is required by user
- ④ High scalability and elasticity

* Disadvantages

- ① Less security

* Examples

AWS, Microsoft Azure, Google Drive, Alibaba cloud, IBM cloud, Oracle cloud

7③ Hybrid Cloud

Combination of public cloud and private cloud.

Here, sensitive data is stored in private cloud and less critical data runs on public cloud.

* Sensitive data → Private

Non-sensitive → Public

* Characteristics

- (i) Flexibility
- (ii) Security
- (iii) Cost-Effective

Lab

* Install "Oracle Virtual Box"

Homework

- Look for the terminologies: -
- (i) Jenkins
 - (ii) DevOps
 - (iii) CI / CD
 - (iv) version control → git
 - (v) docker

(i) Jenkins :- Automation tool used in DevOps

(ii) DevOps :- Development (Dev) + Operations (Ops)
Improve collaboration b/w developers and system admins → planning, development, testing, deployment

(iii) CI / CD :-

CI → Continuous Integration

CD → Continuous Delivery / Deployment

(iv) Git :- Version control system

→ keeps track of code changes

→ Allows multiple developers to work together

→ We can go back to older versions

(v) Docker :- Containerization tool

→ Packages our application with all its dependencies

→ Runs the app the same way on every system

(vi) Containers :- Lightweight environment for running apps.

DevOps and Cloud
building, testing and deploying applications efficiently

16/11/2024

Difference b/w Public cloud and Private cloud	
Public cloud	Private cloud
- owned & managed by 3rd party cloud providers - shared by multiple organizations - Low cost - Low security	- owned & managed by single organization - shared by single organization - High cost - High security

> Multi-Cloud

It uses 2 or more cloud service providers from a number of different computing vendors.

* Purpose : - Reduce risks

Cloud platforms provide a variety of services and resources over the internet, allowing users ~~and~~ access and utilize computing power, storage, db, networking,

* Types difference
* Types services

* What is virtual box

* Installation steps

* gcc compiler

* C programs.

16/11/2026
Friday

Difference b/w cloud and private

public cloud	private cloud
- owned & managed by 3rd party cloud providers - shared by multiple organization - Low cost - Low security	- owned & managed by single organization - shared by single organization - High cost - High security

7 Multi-Cloud

It uses 2 or more cloud service providers from a number of different computing vendors.

* Purpose : - Reduce risks

Cloud platforms provide a variety of services and resources over the internet, allowing users ~~and~~ access and utilize computing power, storage, db, networking,

* Types difference
* Types services

* What is virtual box

* Installation steps

* gcc compiler

* C programs.

* Disaster recovery and backup solutions:
backup and recovered the system

> Cloud Services

Computing services delivered over the internet.

Importance :- agility (ability to move quickly)

- scalability

- cost-effectiveness

- ① IaaS (Infrastructure as a Service)
- ② PaaS (Platform as a Service)
- ③ SaaS (Software as a Service)

> IaaS :- provides virtualized computing resources over the internet.

- * servers, storage & networking infrastructure

- * Benefits :-

- scalability, flexibility, cost-effectiveness

Eg :- AWS EC2, Microsoft Azure

Virtual Machines, Google Compute Engine

→ Instance :- allows user to run virtual servers.

- used to host website & web applications
- for running s/w applications

Uses of IaaS

① Development and Testing environment
used to create scalable develop

→ hosting dynamic websites

② Web hosting and application hosting
also host system like system finally down host 24/7mb.

③ Disaster recovery and backup solutions
disaster recovery, data backup system, systems
secured 24/7mb.

④ PaaS

④ Virtual Data Center

AWS EC2 it can replace
physical vendors with
cloud infrastructure.